

Employee Attrition Analysis

Project Overview :

Employee attrition is one of the most significant challenges faced by organizations, as it affects productivity, recruitment costs, employee morale, and overall business performance. This project analyzes an HR employee dataset to identify the key factors contributing to employee attrition using Tableau.

The analysis explores employee demographics, job satisfaction, overtime, department, education field, business travel, and age distribution to uncover meaningful patterns. Interactive dashboards and visualizations were created to help HR professionals understand workforce trends and support data-driven decision-making.

The primary objective of this project is to identify areas with high employee turnover and provide actionable insights that can help organizations improve employee retention and workforce management.

Dataset Information :

The dataset used in this project contains employee records from an HR department and is commonly used for employee attrition analysis. It includes demographic information, job-related attributes, compensation details, satisfaction ratings, and work-related factors that influence employee turnover.

Dataset Summary :

Attribute	Value
Total Records	1,470
Total Features	35
Target Variable	Attrition (Yes / No)
Dataset Type	Human Resources (HR) Employee Data
Key Features	

The dataset includes information such as:

Employee Age

Gender

Department

Job Role
Education Field
Business Travel
Monthly Income
Job Satisfaction
Environment Satisfaction
Work-Life Balance
Overtime Status
Years at Company
Total Working Years
Distance From Home
Performance Rating
Attrition Status

Objective of the Dataset :

The primary objective of this dataset is to analyze employee characteristics and workplace factors that contribute to employee attrition. By identifying patterns associated with employees leaving the organization, businesses can design effective retention strategies, improve employee satisfaction, and reduce turnover.

Data Cleaning :

Before performing the analysis, the dataset was carefully inspected and cleaned to improve data quality and ensure accurate results.

Data Cleaning Process :

The following preprocessing steps were carried out:

Loaded the dataset into Python using the Pandas library.

Examined the dataset structure, data types, and summary statistics.

Checked for missing values across all columns.

Identified duplicate records and removed them where necessary.

Verified categorical variables to ensure consistent values and correct category labels.

Reviewed numerical columns for invalid or unusual values.

Identified constant columns with no analytical value and excluded them from further analysis.

Verified data integrity before proceeding with exploratory data analysis.

Exported the cleaned dataset for use in SQL and Tableau.

Libraries Used :

The data cleaning process was performed using the following Python libraries:

Pandas

NumPy

Outcome

After cleaning, the dataset was consistent, reliable, and ready for exploratory data analysis, SQL querying, dashboard development, and visualization in Tableau. These preprocessing steps ensured that the insights generated from the analysis were based on high-quality data.

Exploratory Data Analysis (EDA) :

Exploratory Data Analysis (EDA) was conducted to examine employee demographics, workplace characteristics, and factors associated with employee attrition. Tableau was used to create interactive visualizations that highlight important trends and patterns within the dataset. The analysis helps identify areas requiring attention and supports data-driven decision-making for improving employee retention.

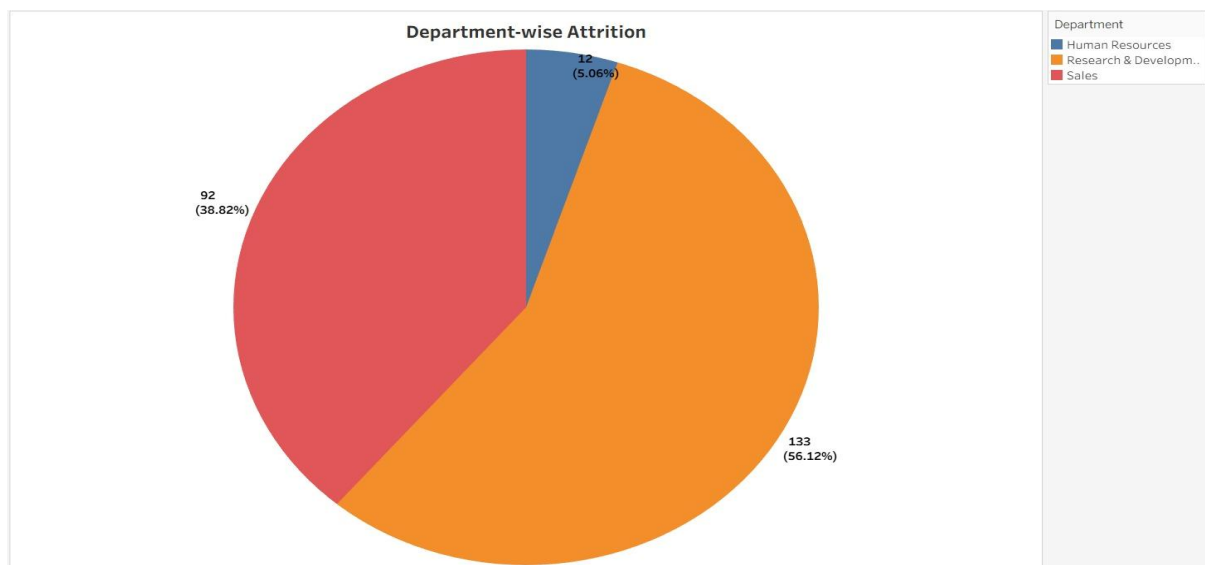
4.1 KPI Summary :

Number Of Employees	Employee Left	Active Employees	Attrition Rate	Avg. Age
1,470	237	1,233	16.12%	37

Explanation :

The KPI dashboard provides a concise overview of the organization's workforce. It summarizes the total number of employees, employees who left the organization, active employees, overall attrition rate, and average employee age. These metrics serve as key performance indicators that help assess the current workforce and establish a foundation for further analysis.

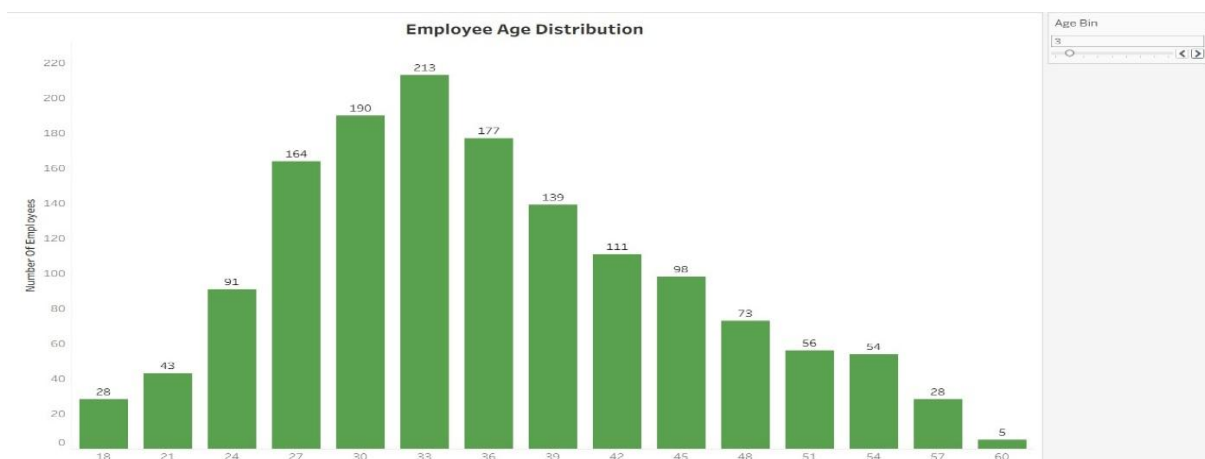
4.2 Department-wise Attrition :



Explanation :

This visualization illustrates the distribution of employee attrition across departments. The Research & Development department contributes the highest number of employee exits, followed by Sales, while Human Resources accounts for the smallest share. These findings suggest that retention initiatives should primarily focus on departments experiencing higher employee turnover.

4.3 Employee Age Distribution :



Explanation :

The histogram displays the distribution of employees across different age groups. The majority of employees fall between the late twenties and late thirties, indicating that the workforce is predominantly composed of mid-career professionals. Comparatively fewer employees belong to the youngest and oldest age groups.

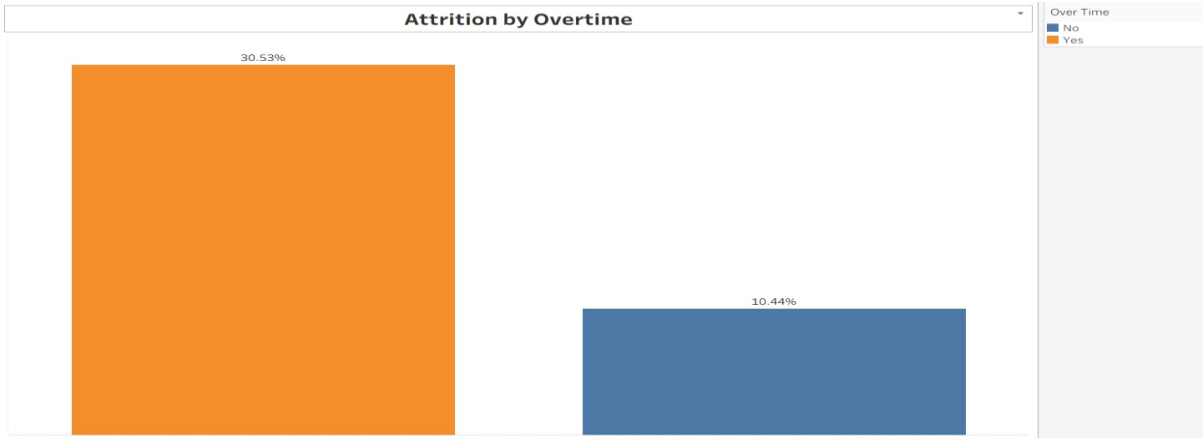
4.4 Job Satisfaction Analysis :



Explanation :

The heatmap compares employee job satisfaction ratings across various job roles. It highlights differences in satisfaction levels throughout the organization and identifies roles with larger employee populations. Such insights help HR teams monitor employee engagement and identify areas that may require improvements in workplace satisfaction.

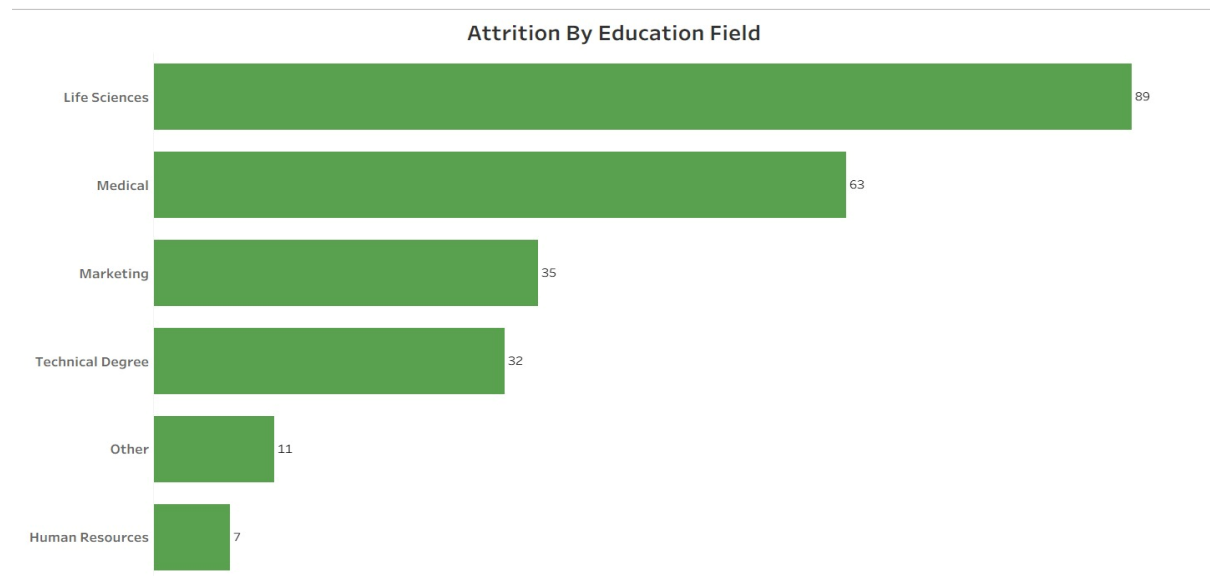
4.5 Attrition by Overtime :



Explanation :

This chart compares attrition rates between employees who work overtime and those who do not. Employees working overtime exhibit a noticeably higher attrition rate, suggesting that extended working hours may contribute to employee turnover. Monitoring overtime policies and maintaining a healthy work-life balance can help improve employee retention.

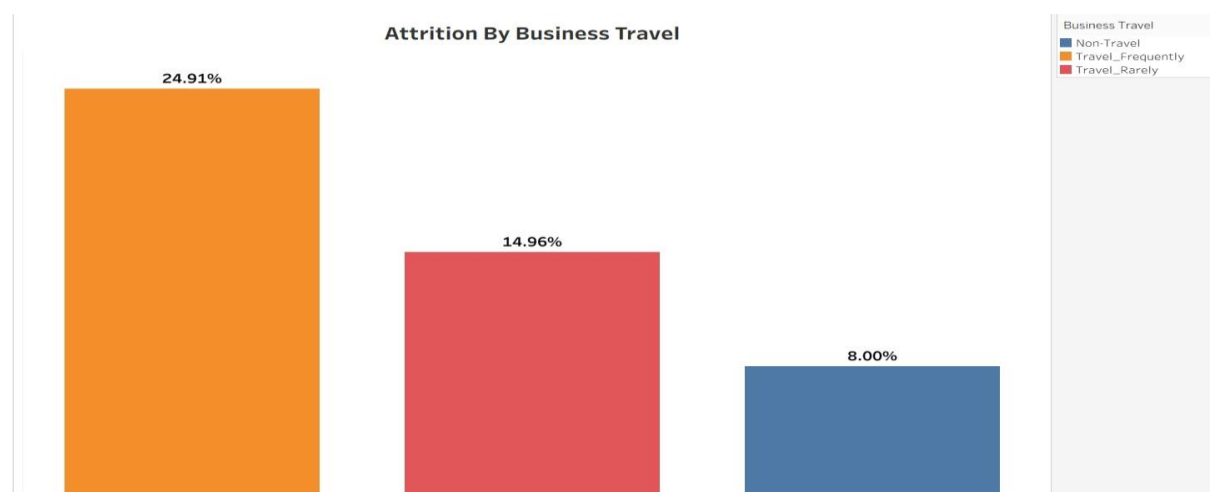
4.6 Attrition by Education Field :



Explanation :

This visualization presents employee attrition across different education fields. Employees with Life Sciences and Medical backgrounds account for the highest number of attrition cases, whereas Human Resources records the lowest. These findings provide insight into workforce composition and highlight educational groups where retention strategies may have greater impact.

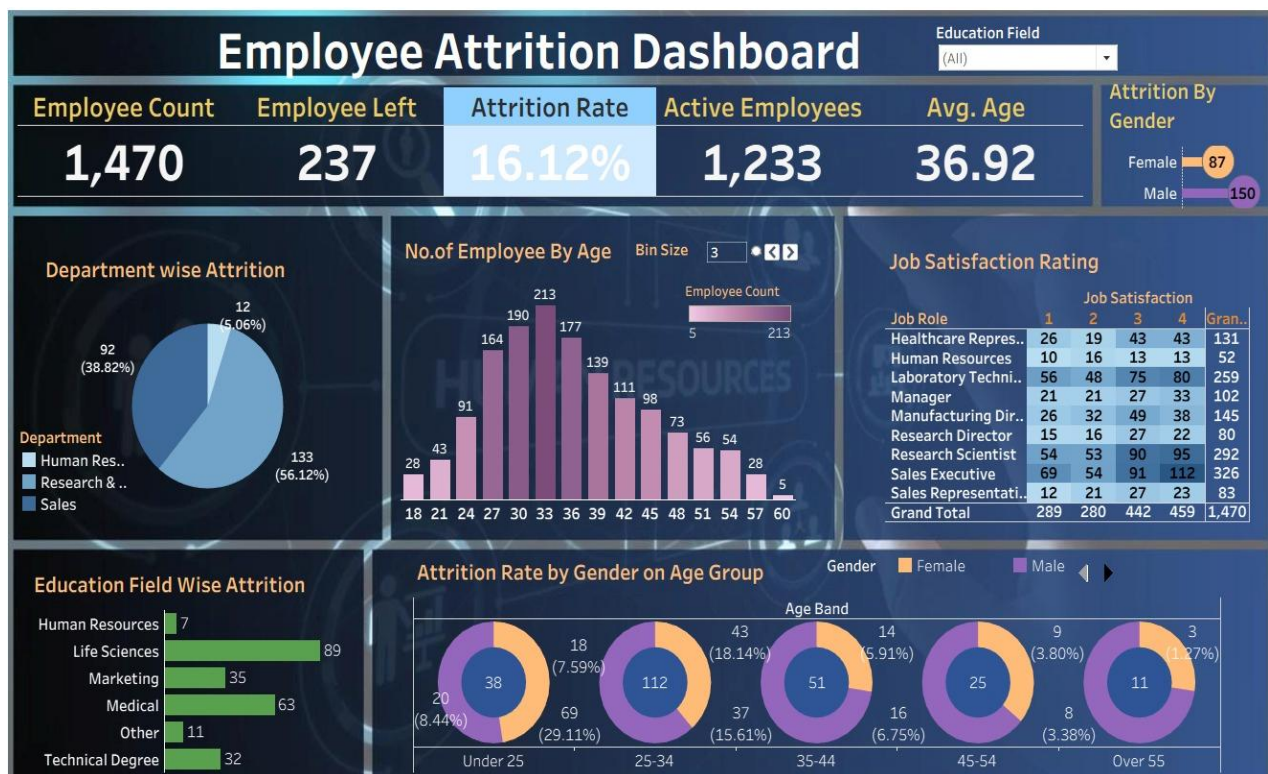
4.7 Attrition by Business Travel :



Explanation :

The chart illustrates the relationship between business travel frequency and employee attrition. Employees who travel frequently experience higher attrition compared to those who travel rarely or do not travel. This suggests that frequent business travel may influence employee satisfaction and long-term retention.

4.8 Interactive Tableau Dashboard



Explanation:

The final Tableau dashboard combines KPIs and multiple interactive visualizations into a single analytical interface. Users can explore employee attrition across departments, education fields, overtime, business travel, job satisfaction, and age distribution. Interactive filters and parameters allow stakeholders to investigate workforce trends dynamically, making the dashboard an effective decision-support tool for HR professionals.

Key Insights :

- The exploratory analysis revealed several important patterns related to employee attrition and workforce characteristics

- The organization has an overall attrition rate of approximately 16.12%, indicating that a notable proportion of employees have left the company.
- The Research & Development department recorded the highest number of employee exits, followed by the Sales department, while Human Resources experienced the lowest attrition.
- The workforce is primarily concentrated between the late twenties and late thirties, representing a predominantly mid-career employee population.
- Employees who work overtime show a significantly higher attrition rate than employees who do not, suggesting that workload and work-life balance influence employee retention.
- Life Sciences and Medical education fields contribute the largest number of employee attrition cases, indicating that turnover is more common among employees from these educational backgrounds.
- Employees who travel frequently for business exhibit higher attrition compared to employees who travel rarely or do not travel.
- Job satisfaction varies across different job roles, highlighting opportunities for targeted employee engagement and satisfaction improvement initiatives.
- The interactive Tableau dashboard enables HR professionals to quickly identify workforce trends and explore employee attrition from multiple business perspectives.

Business Recommendations :

- Based on the analysis, the following recommendations can help improve employee retention and workforce management.
- Prioritize employee retention programs within the Research & Development and Sales departments, where attrition is highest.
- Monitor employee overtime and encourage a healthier work-life balance to reduce turnover associated with extended working hours.
- Conduct regular employee satisfaction surveys to identify workplace concerns before they lead to employee resignation.
- Review business travel policies and provide additional support for employees who travel frequently.
- Develop career growth opportunities and employee engagement programs, particularly for departments and education groups with higher attrition.
- Use HR analytics dashboards to continuously monitor workforce trends and evaluate the effectiveness of retention strategies.
- Implement periodic performance reviews and employee feedback sessions to strengthen communication between employees and management.

Conclusion :

This project successfully demonstrates an end-to-end employee attrition analysis using Python, SQL, and Tableau. The dataset was cleaned and explored to understand workforce characteristics and identify factors associated with employee turnover. Business-oriented SQL queries were used to answer key analytical questions, while Tableau dashboards transformed the data into interactive visualizations for effective decision-making.

The analysis identified important relationships between attrition and factors such as department, overtime, business travel, education field, age distribution, and job satisfaction. These insights can assist HR professionals in designing targeted retention strategies and improving employee satisfaction.

Overall, this project highlights the value of data analytics and business intelligence in supporting informed HR decisions and demonstrates a practical workflow for transforming raw data into meaningful business insights.